A One-Dimensional Obstruction to the Literal Asymmetric Strong Ekeland Extension

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Status

Scoped counterexample, likely valid. This packet refutes the literal unqualified extension of the displayed quasi-pseudometric Suzuki-style strong Ekeland condition to arbitrary asymmetric normed spaces. It does not rule out more carefully formulated asymmetric analogues under additional separation, compactness, bicompleteness, or one-sided strong-minimum hypotheses.

Source Question

The source paper is S. Cobzas, *The strong Ekeland variational principle in quasi-pseudometric spaces*, arXiv:2402.06657, published in *Mathematics* 12 (2024), 471 [1]. Its conclusion asks whether the metric converse compactness characterization has an asymmetric analogue and whether Suzuki's Banach-space/reflexivity theorems can be extended to asymmetric normed spaces.

4. Conclusions

We have proved (Theorem 3.16) a version of strong Ekeland in a quasi-pseudometric space (X, d) having d^s-compact $\bar{d}$-bounded sets. As it was shown by Suzuki [36], in a metric space X the validity of strong EkVP is equivalent to the fact all closed bounded subsets of X are compact. I do not know whether a similar result holds in the asymmetric case - a question that deserves further investigation.

A notion of reflexivity of normed spaces was also considered in the asymmetric case (see [17] or [7, Section 2.5.6]), but in a more complicated way than in the classical one. The extension of Theorems 2.8 and 2.10 to the asymmetric case could be another theme of reflection.

References

The literal condition used below is the displayed quasi-pseudometric analogue of Suzuki's condition in the source, equation (3.16):

The analogs of the conditions (2.5) in the quasi-pseudometric case are:

$$
\begin{aligned}
&\text{(i)}\quad f(z) + \lambda\rho(z, x_0) \le f(x_0)\,;\\
&\text{(ii)}\quad f(y) = f(z) \quad \text{for all} \quad y \in S_\lambda(z);\\
&\text{(iii)}\quad f(z) < f(x) + \lambda\rho(x, z) \quad \text{for all} \quad x \in X \setminus S_\lambda(z)\,;\\
&\text{(iv)}\quad f(x_n) + \lambda\rho(z, x_n) \to f(z) \quad \Rightarrow \quad \lim_{n\to\infty} d(x_n, z) = 0\\
&\qquad\qquad \text{for every sequence } (x_n) \text{ in } X.
\end{aligned}
\tag{3.16}
$$

The quasi-pseudometric analog of Theorem 2.6 is the following.

Theorem 3.16. *Let (X, d) be a quasi-pseudometric space such that every $\bar{d}$-bounded se-*

In the notation of this packet, $S_\lambda(z) = \{y : f(y) + \lambda d(y, z) \le f(z)\}$.

Idea of the Proof

The obstruction is purely directional. In the half-line asymmetric norm, moving to the left from z has zero forward distance $d(z, \cdot)$, but positive reverse distance $d(\cdot, z)$. The source's displayed strong condition asks a minimizing sequence with vanishing forward perturbation from z to converge back to z in the reverse direction. The constant function exposes this mismatch immediately: the sequence constantly equal to $z - 1$ has zero forward perturbation but stays reverse distance 1 from z.

Counterexample

Theorem 1. *Let $X = \mathbb{R}$ and $q(t) = \max\{t, 0\}$. Let*

$$d(x, y) = q(y - x) = \max\{y - x, 0\}.$$

Then q is an asymmetric norm and d is its associated quasi-metric. For the constant function $f \equiv 0$, for every $x_0 \in X$ and every $\lambda > 0$, there is no point $z \in X$ satisfying the literal displayed conditions (3.16) of [1], with $\rho = d$.

Proof. The function q is positively homogeneous for nonnegative scalars and subadditive because

$$\max\{s + t, 0\} \le \max\{s, 0\} + \max\{t, 0\}.$$

Moreover $q(t) = q(-t) = 0$ implies $t = 0$. Thus q is a standard asymmetric norm in the usual T_0 sense, and

$$d(x, y) = q(y - x)$$

is a quasi-metric.

The constant function $f \equiv 0$ is proper, bounded below, and d-lower semicontinuous. Suppose that z satisfies the displayed conditions for some $x_0 \in \mathbb{R}$ and $\lambda > 0$.

Condition (i) gives

$$0 + \lambda d(z, x_0) \le 0,$$

so $d(z, x_0) = q(x_0 - z) = 0$. Hence $x_0 - z \le 0$, i.e. $z \ge x_0$.

For this constant function,

$$S_\lambda(z) = \{y : \lambda d(y, z) \le 0\} = \{y : q(z - y) = 0\} = [z, \infty).$$

Thus condition (ii) is automatic, because f is constant on $S_\lambda(z)$. If $x \notin S_\lambda(z)$, then $x < z$, and

$$f(z) = 0 < \lambda(z - x) = f(x) + \lambda d(x, z),$$

so condition (iii) is also automatic.

It remains to check the strong-minimum condition. Let $x_n = z - 1$ for every n. Then

$$f(x_n) + \lambda d(z, x_n) = 0 + \lambda q((z - 1) - z) = \lambda q(-1) = 0 = f(z).$$

So the premise of condition (iv) is satisfied. But

$$d(x_n, z) = q(z - (z - 1)) = q(1) = 1$$

for all n, and therefore $d(x_n, z) \not\to 0$. This contradicts condition (iv). Since the argument applies to every possible z, no required point exists. □

Verification Notes

No computational verification is needed. The example is one-dimensional and the proof reduces the four displayed conditions to elementary inequalities. The only convention used is the standard quasi-metric associated to an asymmetric norm, $d(x, y) = q(y - x)$. With this convention, the obstruction is exactly the mismatch between $d(z, x_n)$ and $d(x_n, z)$ in condition (iv).

This packet intentionally targets the literal displayed condition in the source. If a future asymmetric-normed-space theorem changes the strong condition to a one-sided convergence requirement, or assumes T_1-type separation/equivalence of the two directions, this example may no longer apply.

Novelty and Search Bounds

Local cheap indexes were searched for arXiv:2402.06657, strong Ekeland, Ekeland variational principle, quasi-metric/quasi-pseudometric spaces, asymmetric normed spaces, Suzuki, reflexivity, Caristi, Takahashi, and completeness. No exact prior packet or attempt was found.

Web searches on 2026-07-18 for the source title, for strong Ekeland plus asymmetric normed spaces and Suzuki/reflexivity, for asymmetric-normed Ekeland variational principles, and for variants involving $\max(t, 0)$ found the source arXiv/MDPI article and related asymmetric Ekeland literature, but no later explicit resolution of this exact one-sided half-line obstruction.

Human Review Recommendation

Accept as a scoped counterexample to the naive literal asymmetric-normed extension of the displayed Suzuki-style condition. Do not count it as a full closure of the source's broader program of finding the right asymmetric reflexivity or compactness theorem.

References

[1] S. Cobzas. *The strong Ekeland variational principle in quasi-pseudometric spaces.* arXiv:2402.06657, 2024. Published in *Mathematics* 12 (2024), 471. DOI: 10.3390/math12030471.

[2] T. Suzuki. *Characterizations of reflexivity and compactness via the strong Ekeland variational principle.* Nonlinear Analysis 72 (2010), no. 5, 2204–2209.